Shivam Khatri

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Summary

- **Mechatronics** Engineering student at the University of **Waterloo** with **2+ years** of software development experience
- Strong foundation in **C++** and building **AI-powered** applications using the Gemini API, Python, and JavaScript
- Curious to learn new things with great teamwork skills, and a lot of experience in working under hard constraints

Technical Skills

Programming Languages: **Python**, SQL, Java, C++, HTML, CSS,embedded systems, Pandas, GenAI

Testing & QA: Software testing, **API** validation, regression testing concepts, structured debugging

Hardware & Design: **SolidWorks**, Fusion 360, AutoCAD, GD&T, Microsoft Office, G-Suite, Arduino, ESP32, VESC motor controllers, IMU (MPU6050), PID control, **sensor integration**, motor control

Experience

Front End Developer — September 2024 – June 2025

STEM Unites

- Increased member engagement by **25%** by redesigning the front end and implementing **optimized** components using JavaScript, HTML, and CSS while using **Git version control** to allow for seamless collaboration
- Improved page load performance by **35%** by refactoring front-end and **back-end code**, debugging performance bottlenecks, and implementing reusable, modular components following software engineering best practices

Firmware Engineer — September 2025 – January 2026

Electrium Mobility

- Bridged a **UART** integration layer between ESP32 and VESC, enabling control of **10+ motor** parameters
- Developed a TFT dashboard displaying real-time telemetry data including speed, voltage, and battery percentage
- Minimized **SPI** bus writes by implementing dirty-flag state detection, redrawing elements only on telemetry state change
- Designed and optimized C/C++ firmware to control motor operation, sensor inputs, and **power systems**

Lead Programmer — September 2022 – April 2025

Vex Robotics

- Increased scoring by **14%** by designing and optimizing motion algorithms using **PID** and iterative testing cycles
- Reduced mechanical integration errors by creating CAD models and technical drawings in Fusion 360
- Won 4 Design Awards by creating a 300+ page documentation file, 3 semi-finalists, and placed 4th overall in Alberta

Head of 3D Printing — September 2021 – June 2025

Green Initiative

- Reduced costs by **22%** by optimizing a in-house 3D printing workflow by reusing recycled plastics as filament
- Increased self-produced filament consistency by **70%** by creating a custom cooling chamber to elevate the filament before entering a self-spooling mechanism essentially reducing fluctuations in filament diameter

Projects

Autonomous Page Navigator/Summarizer | *JavaScript, Gemini API, DOM Manipulation* — Sepetember 2025

- Created a Google Chrome extension connected to the **Gemini API** which had an AI summarizer tool capable of parsing a long page in 2 seconds and creating a summary for the user to read
- Reduced website navigation time to 3 seconds by developing a feature where the user sends a prompt to the Gemini API, which then parses the information received in the **JSON** response, and executes actions sequentially using DOM Manipulation, eventually rerouting the user to the target page in only a few seconds

StabiliKnee - Predictive Orthosis System | *C++, Arduino, Embedded Systems* — February 2026

- Designed a predictive knee stabilization system using **IMU**-based motion sensing, reducing simulated patella dislocations by **70%** and earning **3rd place** at MakeUofT, one of Canada's largest hardware hackathons
- Developed Arduino C++ firmware which used kinematic analysis read data and compare it against a threshold range and detect dangerous movements **300ms** within happening and trigger a corrective motor response

JP Morgan - Software Engineering Simulation | *Java, RESTful API*

- Designed and implemented **RESTful** APIs in Spring Boot to validate and process financial transactions
- Implemented object-oriented programming practices such as inheritance to create different account types and databases.

Education

University of Waterloo

*Honors Mechatronics Engineering, Co-op Program–**3.7 GPA***